



ROBERT H. SMITH SCHOOL OF BUSINESS

NFL Offensive Strategy Analysis

How teams adjust their offense when the score changes.

September 10, 2026

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Football is a game where teams have to constantly adjust their strategy based on what is happening on the field. One factor that can have a major impact on offensive decision-making is the score. A team that is leading may approach its offense differently than a team that is trying to come back from a deficit. As a team falls further behind, it may rely more heavily on passing in an attempt to score and close the gap. This raises the question of **how offensive strategy changes when NFL teams are winning versus losing**. In this project, I analyze NFL play-by-play data to examine how teams change their use of passing and rushing plays depending on the score of the game.

How Does Offensive Strategy Change When NFL Teams Are Winning vs. Losing?

Introduction

The primary research question for this analysis is: **How does offensive strategy change when NFL teams are winning versus losing?** More specifically, this project examines how play selection changes when teams are trailing by 7 or more points compared with when they are leading or tied. The analysis focuses on the percentage of offensive plays that are passes versus runs. It also examines average yards gained and how play selection changes throughout the game when a team is trailing by 7 or more points.

Using NFL play-by-play data from the 2021 through 2024 seasons, this analysis identifies patterns in offensive strategy based on the game situation. The results provide insight into how teams adjust their offensive approach as the score changes and as the game progresses.

Data and Methodology

The analysis used NFL play-by-play data from the **2021, 2022, 2023, and 2024 seasons**. The 2024 dataset contained **49,492 plays and 372 variables**. For the main analysis, the data was filtered to include only plays classified as either a **run or pass**, resulting in **35,050 offensive plays** in 2024.

The analysis categorized each offensive play according to the team's score differential at the time of the play. Five game-situation categories were created:

- Leading: Team is ahead by at least 1 point
- Tied: Score is tied
- Trailing: Team is behind by 1-6 points
- Trailing by 7+: Team is behind by 7-13 points
- Trailing by 14+: Team is behind by 14 or more points

The percentage of pass and run plays was then calculated within each game-situation category. This allowed the analysis to compare the offensive play selection as the size of the deficit changed.

In addition to play selection, average **yards gained** and **expected points added (EPA)** were calculated for both passing and rushing plays across the different game situations. Finally, the analysis focused specifically on teams trailing by 7 or more points and compared their pass and run percentages across the four quarters of the game.

The analysis was conducted using **R** and the **nflreadr** package to load and analyze NFL play-by-play data.

Results

Offensive Play Selection by Game Situation

The first part of the analysis examined how the percentage of passing and rushing plays changed based on the team's score differential.

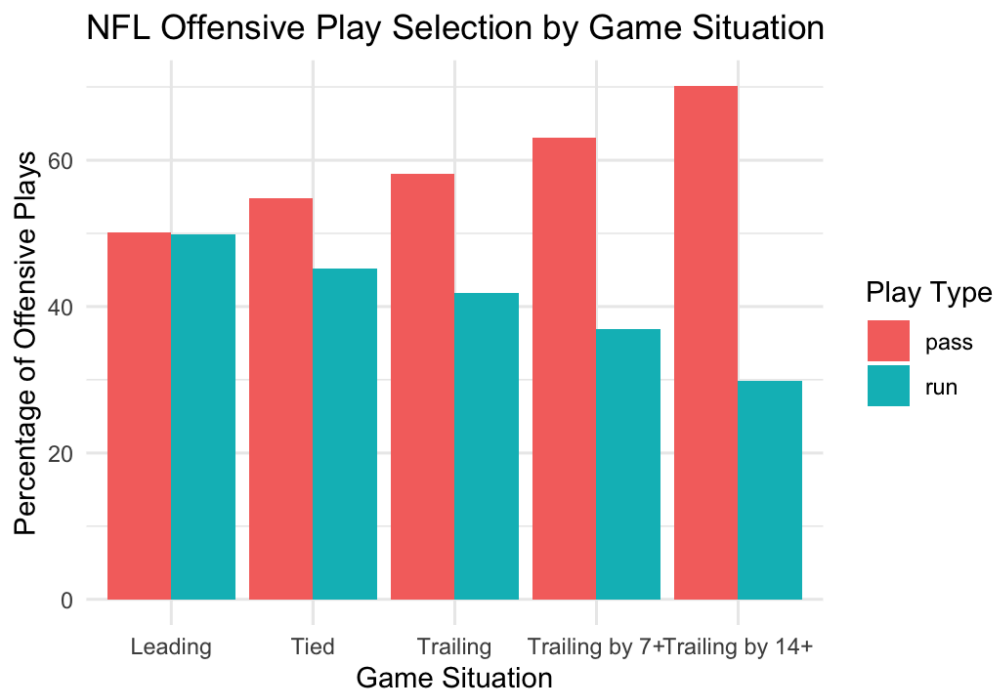


Figure 1. NFL Offensive Play Selection by Game Situation

The results show a clear increase in passing as teams move from leading to trailing by larger margins. When teams were leading, **50.09%** of their offensive plays were passes and **49.91%** were runs. When the score was tied, the passing percentage increased to **54.81%**. Teams trailing by 1-6 points passed on **58.20%** of their offensive plays.

The difference became larger when teams were facing bigger deficits. Teams trailing by 7-13 points passed on **63.04%** of their offensive plays, compared with 36.96% runs. When teams were trailing by 14 or more points, passing accounted for **70.13%** of offensive plays, while rushing accounted for only 29.87%.

Overall, the analysis shows that passing became increasingly common as the size of the deficit increased. Comparing teams that were leading with teams trailing by 14 or more points, the percentage of passing plays increased by **20.04 percentage points**.

Offensive Efficiency by Game Situation

The analysis also examined the average yards gained on passing and rushing plays across each game situation.

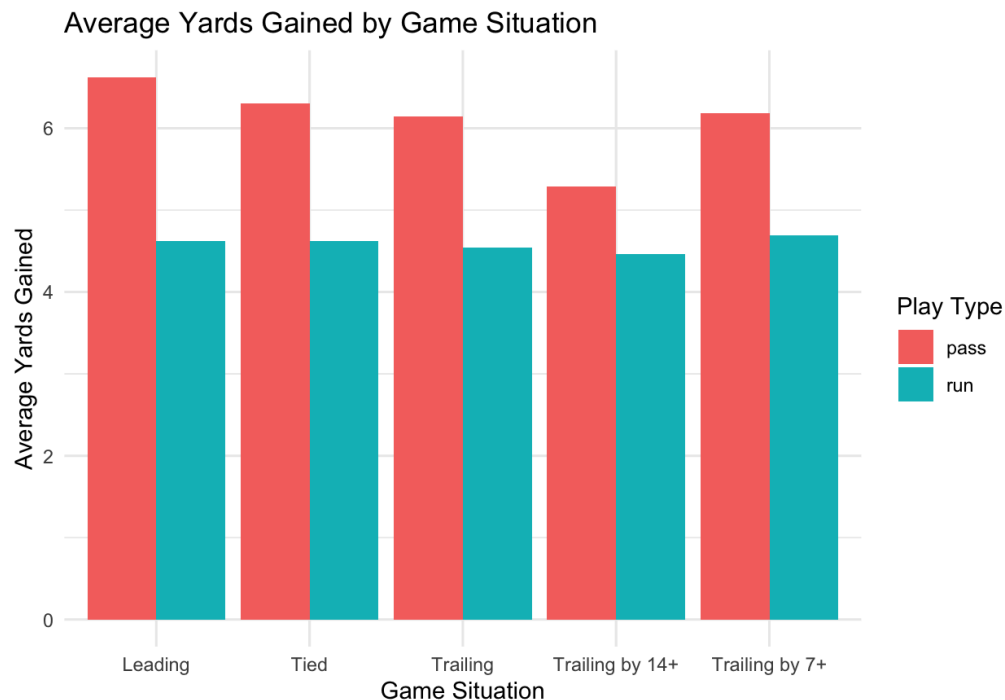


Figure 2. Average Yards Gained by Game Situation

When teams were leading, passing plays averaged **6.62 yards**, while rushing plays averaged **4.62 yards**. When the score was tied, passing averaged 6.30 yards and rushing averaged 4.62 yards.

For teams trailing by 7 or more points, passing plays averaged **6.18 yards**, while rushing plays averaged **4.69 yards**. However, when teams were trailing by 14 or more points, average passing yards dropped to **5.29 yards**, while rushing plays averaged **4.46 yards**.

These results show that although teams passed more frequently when trailing, increased passing volume did not necessarily correspond with higher average yards per passing play. In particular, teams trailing by 14 or more points had the lowest average passing yards among the game situations examined.

The EPA results also provide additional context. Passing EPA was **0.087** when leading but decreased to **0.020** when trailing by 7 or more points and **-0.122** when trailing by 14 or more points. This suggests that the increased reliance on passing while trailing did not automatically result in more efficient offensive plays.

Offensive Strategy When Trailing by 7+

The next part of the analysis examined whether teams changed their offensive strategy as the game progressed when they were trailing by 7 or more points.



Figure 3. Offensive Play Selection When Trailing by 7+

In the first quarter, teams trailing by 7 or more points passed on **53.81%** of their offensive plays and ran on 46.19%. Passing increased to **60.77%** in the second quarter.

The passing percentage decreased slightly to **56.91%** in the third quarter before increasing substantially in the fourth quarter. By the fourth quarter, teams trailing by 7 or more points passed on **77.63%** of their offensive plays and ran on only **22.37%**.

The difference between the first and fourth quarters represents a **23.82 percentage-point increase in passing**. This is the largest shift observed in the quarter-by-quarter analysis.

The results suggest that the urgency to pass becomes particularly pronounced later in the game when a team remains behind by 7 or more points. Rather than maintaining a relatively balanced offensive approach, teams increasingly rely on passing as the game approaches its conclusion.

Multi-Season Results

To determine whether the 2024 findings were isolated to one season, the analysis also examined teams trailing by 7 or more points across four seasons.

Season	Pass Percentage When Trailing by 7+
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2021	63.69%
2022	62.97%
2023	63.09%
2024	63.04%

The results were remarkably similar across all four seasons. Passing accounted for approximately **63% of offensive plays when teams were trailing by 7 or more points in each season**. The percentages ranged from 62.97% in 2022 to 63.69% in 2021.

The consistency across seasons provides additional context for the 2024 findings. The increased reliance on passing when trailing by 7 or more points was not limited to the 2024 season in this analysis.

Discussion

The findings demonstrate a relationship between game situation and offensive play selection. As the size of a team's deficit increased, the percentage of passing plays also increased. Teams that were leading used a nearly even balance of passes and runs, while teams trailing by larger margins relied increasingly on the passing game.

The largest difference occurred between teams that were leading and teams trailing by 14 or more points. Passing increased from **50.09% to 70.13%**, representing a 20.04 percentage-point increase.

The quarter-by-quarter analysis also demonstrated that the score situation becomes increasingly important as the game progresses. When trailing by 7 or more points, teams passed on 53.81% of their plays in the first quarter but 77.63% in the fourth quarter. This suggests that teams facing a deficit may become increasingly pass-oriented as they have less time remaining to recover.

However, the efficiency results show that passing more often does not necessarily mean passing more effectively. Teams trailing by 14 or more points had an average of only 5.29 yards per pass, compared with 6.62 yards when leading. Their passing EPA was also negative at -0.122. This provides an important distinction between **offensive strategy and offensive efficiency**. Teams may increase passing because the game situation requires them to take a more aggressive approach, even if that approach becomes more difficult to execute.

The consistency across the 2021-2024 seasons also supports the idea that this pattern is not unique to one season. Across all four seasons, teams trailing by 7 or more points passed on approximately 63% of their offensive plays.

Limitations

There are several limitations to this analysis. First, the analysis focuses on play selection and does not account for every factor that can influence an offensive decision. Factors such as time remaining, down

and distance, field position, team personnel, injuries, weather, and individual team offensive philosophies may affect whether a team chooses to run or pass.

Second, the analysis groups teams based primarily on score differential. Two teams could both be trailing by 7 points but face very different circumstances depending on the quarter, time remaining, field position, or down and distance.

Additionally, the analysis examines relationships between game situations and play selection rather than establishing that the score differential itself directly causes teams to change their strategy.

Conclusion

This analysis examined how NFL offensive strategy changes based on whether teams are leading, tied, or trailing. The results show a clear increase in passing as teams fall further behind. Passing accounts for 50.09% of offensive plays when teams were leading, compared with 63.04% when trailing 7-13 points and 70.13% when trailing by 14 or more points.

The analysis also found that teams trailing by 7 or more points became substantially more pass-oriented as the game progressed. Passing increased from 53.81% of offensive plays in the first quarter to 77.63% in the fourth quarter.

At the same time, the efficiency results demonstrate that increased passing volume did not necessarily produce better results. Teams trailing by 14 or more points had lower average passing yards and negative average EPA compared with teams in less difficult game situations.

Overall, the findings show that **NFL offensive strategy changes based on the score and timing of the game**. Teams facing larger deficits increasingly rely on the passing game, particularly late in the game, as they attempt to overcome the deficit.

References

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